Growth Rate :

$K \log n < n < n \log n < n^2 < n^3 < 2^n \ldots$

- For O → look upward : if $f(n) = n$ then $n \in O(n)$, $n \in O(n^2)$, $n \in O(n^3)$ ...
- For Ω → look downward : if $f(n) = n^3$ then $n^3 \in \Omega(n^3)$, $n^3 \in \Omega(n^2)$, $n^3 \in \Omega(n)$.
- For Θ → only same level : if $f(n) = 2n^2 + 2$ then $f(n) \in \Theta(n^2)$.

1. $2n^2 + 2 \in O(n)$.
2. $2n^2 + 2 \notin \Omega(n^2)$.
3. $2n^2 + 2 \in \Omega(n)$.
4. $2n^2 + 2 \in \Theta(n)$.
5. $2n + 2 \in O(n^2)$.
6. $2n + 2 \in \Omega(n^2)$.
7. $2n + 2 \in O(n^2)$.

8. $3n+2 \in O(n)$.

9. $10n^2+4n+2 \in O(n)$.

10. $10n^2+9 \notin O(n)$.

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1. $\{2n^2+2 \in O(n)\}$

c	n_0	n	$2n^2+2 \leq c \cdot n$	TRUE/FALSE
1	1	1	$4 \leq 1$	False
2	1	1	$4 \leq 2$	False
3	1	1	$4 \leq 3$	False
4	1	1	$4 \leq 4$	True
4	1	2	$10 \leq 8$	True
4	1	3	$20 \leq 12$	False

2. $2n+2 \notin \Omega(n^2)$.

c	n_0	n	$2n+2 \geq c \cdot n$	TRUE/FALSE
1	1	1	$4 \geq 1$	True
1	1	2	$34 \geq 4$	True
1	1	3	$164 \geq 9$	True

3. $2n^3 + 2 \in \Omega(n)$.

c	n_0	n	$2n^3 + 2 \geq c \cdot n$	TRUE/FALSE
1	1	1	$4 \geq 1$	True
1	1	2	$18 \geq 1$	True
1	1	3	$56 \geq 3$	True for all n

5. $2n+2 \in O(n^2)$

c	n_0	n	$2n+2 \leq c \cdot n^2$	TRUE/FALSE
4	1	1	$4 \leq 4$	True
4	1	2	$6 \leq 16$	True
4	1	3	$8 \leq 36$	True for all n

6. $2n^4 + 2 \in \Omega(n^2)$.

c	n_0	n	$2n^4 + 2 \geq c \cdot n^2$	TRUE/FALSE
1	1	1	$4 \geq 1$	True
1	1	2	$34 \geq 4$	True
1	1	3	$164 \geq 9$	True for all n

7. $2n^4 + 2 \in O(n^2)$.

c	n_0	n	$2n^4+2 \leq c \cdot n^2$	TRUE/FALSE
1	1	1	$4 \leq 1$	False
4	1	1	$4 \leq 4$	True
4	1	2	$34 \leq 16$	False
4	1	3	$164 \leq 36$	False.

8. $3n+2 \in O(n)$.

c	n_0	n	$3n+2 \leq c \cdot n$	TRUE/FALSE.
1	1	1	$5 \leq 1$	False
4	1	1	$5 \leq 4$	False
4	2	2	$8 \leq 8$	True
4	2	3	$11 \leq 12$	True
4	2	4	$14 \leq 16$	True for all $n \geq 2$

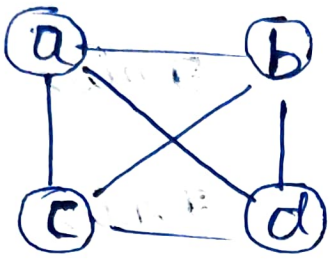
9. $10n^2 + 4n + 2 \in O(n)$

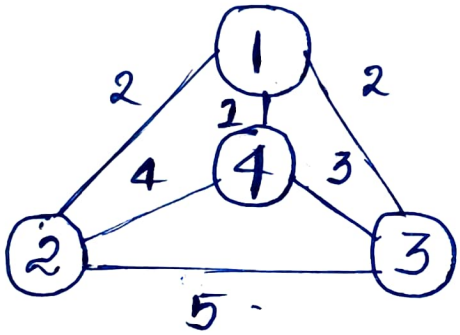
C	n_0	n	$10n^2 + 4n + 2 \leq C \cdot n$	TRUE/FALSE
16	1	1	$16 \leq 16$	True
16	1	2	$50 \leq 32$	False
16	1	3	$104 \leq 48$	False
100	1	10	$1042 \leq 1000$	False.

10. $10n^2 + 9 \notin O(n)$.

C	n_0	n	$10n^2 + 9 \leq C \cdot n$.	TRUE/FALSE.
19	1	1	$19 \leq 19$	True
19	1	2	$49 \leq 38$	False
19	1	3	$99 \leq 57$	False
100	1	10	$1009 \leq 1000$	False; for all $n \geq 10$

Travelling salesman Problem





$1 \xrightarrow{2} 2 \xrightarrow{5} 3 \xrightarrow{3} 4 \xrightarrow{1} 1 \Rightarrow 11$

$1 \xrightarrow{2} 2 \xrightarrow{4} 4 \xrightarrow{3} 3 \xrightarrow{2} 1 \Rightarrow 11$

$1 \xrightarrow{2} 3 \xrightarrow{3} 4 \xrightarrow{4} 2 \xrightarrow{2} 1 \Rightarrow 11$

$1 \xrightarrow{2} 3 \xrightarrow{5} 2 \xrightarrow{4} 4 \xrightarrow{1} 1 \Rightarrow 12$

~~3 2 4~~

$1 \xrightarrow{1} 4 \xrightarrow{4} 2 \xrightarrow{5} 3 \xrightarrow{2} 1 \Rightarrow 12$

$1 \xrightarrow{1} 4 \xrightarrow{3} 3 \xrightarrow{5} 2 \xrightarrow{2} 1 \Rightarrow 11$